

CLINIC-CLUSTER MAPPER

AI-Powered Healthcare Accessibility & Telemedicine Hub Optimization

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IMPORTANT DISCLAIMER: This is an academic decision-support prototype. All data used is synthetic/demo data. This system is NOT a medical diagnosis tool and should NOT be used for actual clinical or policy decisions without expert review. All weights and thresholds are modeling assumptions, NOT medical standards.

1. Executive Summary

This report analyzes healthcare accessibility across 500 communities with a total population of 3,388,259. The average healthcare equity score is 54.6/100. 449 communities (89.8%) were identified as underserved. 3 telemedicine hub locations have been recommended.

2. Dataset Statistics

Metric	Value
Communities	500
Total Population	3,388,259
Healthcare Facilities	35
Specialist Centers	18
Avg Distance to Clinic	25.2 km
Avg Distance to Specialist	83.8 km
Avg Connectivity	0.44
Avg Road Accessibility	0.53

3. Cluster Analysis

K-Means clustering was performed with K=3. Silhouette Score: 0.3222. Inertia: 2903.59.

Cluster	Label	Size	Avg Equity Score
0	Poor Connectivity	167	N/A
1	Vulnerable / High Demand	43	N/A
2	Average Access	290	N/A

4. Healthcare Equity Analysis

Healthcare equity scores range from 14.7 to 93.4 with a mean of 54.6 and median of 54.0. The score is a weighted composite of specialist distance (25%), clinic distance (15%), vulnerability index (20%), demand ratio (15%), connectivity (15%), and road access (10%).

Note: These weights are modeling assumptions for academic analysis, NOT official medical standards.

5. Underserved Areas

449 communities were identified as underserved. Of these, 20 are classified as Critical priority and 85 as High priority.

Village	Population	Priority	Equity Score
Harikhedi (SYNTHETIC)	1530	Medium	48.0
Chandangarh (SYNTHETIC)	2859	Medium	43.2
Laxmangarh (SYNTHETIC)	1354	Medium	71.2
Gopalgaon (SYNTHETIC)	3870	Medium	55.8

Bhimgaon (SYNTHETIC)	1244	Medium	58.8
Nandkhedi (SYNTHETIC)	3233	Medium	63.2
Gopalpura (SYNTHETIC)	466	Critical	23.8
Rajpur (SYNTHETIC)	1509	Medium	67.0
Sultangarh (SYNTHETIC)	1650	Medium	55.4
Nandpur (SYNTHETIC)	10894	Medium	79.7

6. Recommended Telemedicine Hub Locations

3 telemedicine hub locations were recommended using population-first optimization.

Hub #	Lat	Lon	Communities	Pop. Covered	Improvement (km)
1	22.6458	78.8599	155	1,178,905	0.00
2	23.3821	77.1451	201	1,449,303	6.51
3	24.6854	78.5467	144	760,051	2.08

Total population covered: 3,388,259

Vulnerable population covered: 1,803,557

Average accessibility improvement: 3.22 km

7. Methodology

This analysis uses K-Means clustering on standardized healthcare accessibility features, a weighted composite equity score, threshold-based underserved detection, and facility-location optimization for hub placement. K-Means++ initialization with iterative refinement is used for hub optimization.

8. Limitations & Data Quality

Key limitations of this analysis:

1. Demo/synthetic data is used - not actual healthcare records.
2. Distances are straight-line (haversine), not actual road distances.
3. Equity score weights are modeling assumptions, not medical standards.
4. The model does not account for political, budgetary, or regulatory constraints.
5. This is an academic prototype and should not be used for actual clinical decisions.
6. No real patient data or PII is used in this system.